
SUMERIAN OCR: DETECTION, RECOGNITION AND TRANSCRIPTION OF CUNEIFORM SIGNS ON SUMERIAN TABLETS

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ABSTRACT

We present a novel approach for optical character recognition (OCR) of Sumerian cuneiform tablets by using an Object Detection framework, YOLO12. In this work, we treat each cuneiform sign as an object and apply a detection-classification pipeline to localize and recognize signs on tablet images. Our system further incorporates a reading order module to sequentially transcribe detected signs into Unicode cuneiform text. Experiments on HeiCuBeDa and MaiCuBeDa datasets demonstrate that this approach improves detection and classification accuracy compared to previous methods, while providing a first step toward automated Sumerian transcription.

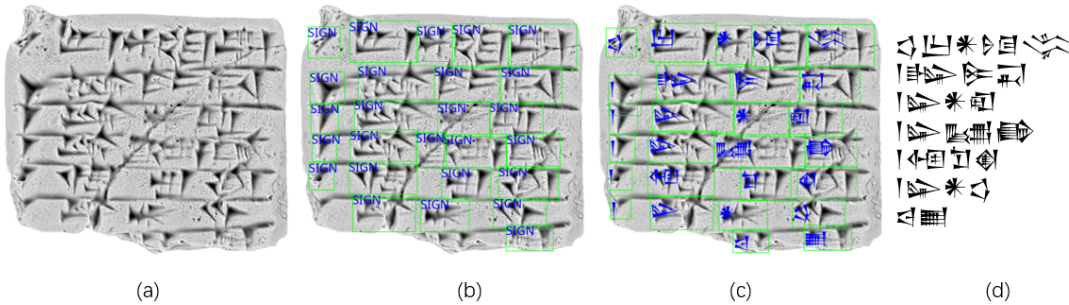


Figure 1: A sample result.

(a) An input image; (b) Sign detection; (c) Sign detection and classification; (d) Transcription.

1 Introduction

Sumerian cuneiform is one of the earliest known writing systems, inscribed on clay tablets over 4,000 years ago. Despite its historical importance, automated reading of cuneiform remains a challenging task. Unlike modern scripts, cuneiform is three-dimensional and highly dependent on lighting and viewing angles, making classical OCR approaches ineffective. Moreover, large-scale annotated datasets are scarce, limiting the applicability of end-to-end recognition models.

Recent advances in computer vision, particularly Object Detection frameworks, provide a promising alternative. By treating each cuneiform sign as a distinct object, modern detection models can be adapted to localize and classify signs on tablet images. Once detected, these signs can be sequentially organized into their natural reading order, enabling automated transcription of Sumerian text.

In this work, we propose a unified pipeline for Sumerian OCR using an Object Detection backbone YOLO12 [6], followed by a reading order module for transcription. We

evaluate our approach on the MaiCuBeDa [3] dataset. Our implementation is publicly available on Github.

1.1 Task description.

Given an input image depicting a Sumerian tablet, we are interested in three types of tasks:

- **Detection:** give the bounding box of each Sumerian sign on a tablet.
- **Detection and Classification:** give the bounding box of each Sumerian sign on a tablet, together with the name of that sign.
- **Transcription:** in Unicode, directly give all the Sumerian text on a tablet sequentially.

Sumerian tablets are recorded either by an photograph, or by "Lineart" drawings before the popularization of photography. In our project, we focus only on photographs (in our case renderings of a 3D model), and do not take into account the Lineart drawings.

1.2 Challenges.

The main challenges of these tasks lie in the following facts:

- **Shortage of high-quality annotated sources:**
Till this day, we still lack large-scale well-annotated image datasets.
For the Detection or Detection and Classification task, Dencker et al. [1] proposed a set of manually labeled bounding boxes, but the number was limited (only 81 images). To accommodate with this shortage, they had to use a much more complex heuristic to put the signs on the image. Only recently did the datasets HeiCuBeDa [2] and MaiCuBeDa [3] appear, with 1070 images annotated with bounding boxes. This is the dataset that we shall use. However, this dataset also has multiple issues, including wrong view name, wrong character name, lots of signs not annotated, etc. See section 3 for details about dataset treatment.
For the Transcription task, Simmons [4] collected a huge dataset of images or Linearts and their corresponding transcription, but the images are not cropped into separate views of a tablet; instead, the 6 views are fused together, making it complicated to effectively train models.
- **Lack of adapted models:**
To our knowledge, there are not yet any language or visual model specifically pretrained on any cuneiform language or image corpus. Popular Large Language Models, such as Gemini 3 and ChatGPT-5, still do not have any grammatical or visual knowledge of Sumerian. Cuneiform scripts are not yet supported by any popular OCR frameworks, such as TrOCR [11] and DeepSeek-OCR [13].

- **Characteristics of the cuneiform script itself:**

The cuneiform script is not flat but 3 dimensional, since it is pressed onto clay or carved onto stone. The information is not conveyed by black and white contrast (like written scripts such as Latin alphabets or Chinese handwriting), but via depth and reflection of light. This makes sign reading particularly challenging, since the same sign may look different under different lighting and reading angles, and one single photo may not capture all information.

2 Related works

Although existing attempts to solve our tasks are limited, some works still provide valuable insights. These include:

- Dencker et al. [1]. They performed the Detection and Classification task. To do with the fact that their annotated bounding boxes were limited, their pipeline was quite complex: they first trained a line detector, then used sign size as heuristics to place known signs onto the image; they then trained iteratively, alternating between aligning the entire transliteration on the image (by minimizing an energy of the word spreading), and to use this as ground truth to further tune the detector.
- Stötzner et al. [7]. They are also the creators of the annotation dataset MaiCuBeDa [3]. They only performed the Detection task. With a RCNN-style model, they trained a wedge detector to get representative points of each sign, then from that deduced the existence of a sign. They also proposed lighting augmentation by applying virtual renderings on 3D models of the tablet, and trained the model on multiple types of images, including photos, VirtualLight renderings and Multi-Scale Integral Invariant (MSII) renderings. They did not, however, try to classify each bounding box to get its sign name.
- Simmons [4]. This was his Computer Vision coursework project at Stanford University. He performed the Transcription task. He proposed a transformer-based image-to-sequence model, consisting of a ViT encoder and an XLM-R decoder. He trained it end-to-end on Lineart drawings.

3 Datasets

3.1 Dataset description.

We use the image dataset called HeiCuBeDa [2], which provides 3D mesh of Sumerian tablets, as well as their MSII rendering and VirtualLight rendering. It also provides transliteration of each tablet and various other meta-data, such as the era of the tablet. We use only the MSII images for our experiments.

We use the MaiCuBeDa [3] annotations on the HeiCuBeDa image set, which provides sign-level and line-level bounding box annotations on 1070 VirtualLight/MSII images. It does not, however, publish bounding boxes on raw photos. We only use the sign-level bounding boxes for our experiments.

3.2 Dataset treatment.

We perform the following treatments on the dataset to make it more usable:

- We discard all images that represent the Left, Right, Top or Bottom side of the tablet, since they contain few annotations.
- We manually fix several cases where the corresponding lines in the csv annotation file have wrong format, or where the annotations are put on the wrong side, or where the sign annotation has typos, or where the bounding box has negative width or height.
- We convert the dataset into YOLO-friendly yaml format.

4 Experiments

4.1 Model Selection.

We choose to use the most recent YOLOv12m [6] model, and to train it under the Ultralytics framework, which offers most of the necessary data augmentation utilities that we need.

YOLOv12 uses an attention-based backbone instead of the older CNN-only design. It introduces a fast Area Attention (A2) module that splits features into simple regions so attention stays efficient and keeps a wide receptive field. It combines A2's into a backbone called Residual Efficient Layer Aggregation (R-ELAN), which stabilizes training with lightweight residual and aggregation layers.

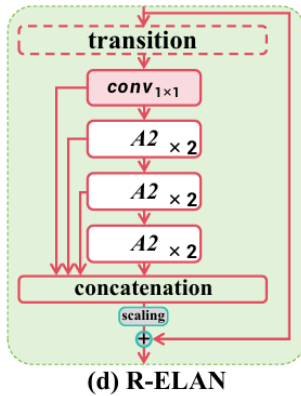


Figure 2: The YOLOv12 architecture.
(Figure from original paper)

4.2 Approach

- To do the Detection and Classification task, we train the YOLOv12m model on the bounding box dataset.

The dataset contains 451 different signs in total. However, many signs are rare and thus hurt performance. We propose to regroup the rarest classes into a single class called "OTHER", keeping only 251 classes, including the one called "OTHER".

- To do the Detection task (without classifying each sign), we train the YOLOv12m model on a simplified dataset where all signs are labeled as "SIGN".
- To do the Transcription task, we simply add a Reading Order Module after doing the Detection and Classification task, which takes in the predicted bounding boxes given by YOLO, and rearrange them according to natural reading order. This contains two steps: first, a Line Cluster step, which uses the DBSCAN algorithm to find horizontal lines by regrouping boxes that share similar y (vertical position) values; second, sort the signs within each line by their horizontal position. The parameter ϵ (same-cluster distance threshold) used in the DBSCAN algorithm is dynamically set to $0.35 \times$ the mean height of all predicted bounding boxes on the image.

We haven't yet implemented column segregation for the cases where the signs are divided into multiple columns. We removed this case from Transcription evaluation.

4.3 Training configuration.

We use 80% of the images as training set and the rest as testing set. Input images are resized to 640×640 . We train with batch size 4 for 70 epochs (the results reported are those of the best epoch), on a single RTX 3060 GPU. The optimizer is the AdamW optimizer with default learning rate ($3.9e-5$). To mitigate overfitting, we perform random data augmentation with the following parameters:

- 7 degrees as maximal rotation,
- $0.1 \times$ side length as maximal translation,
- $\pm 50\%$ as maximal scaling,
- 7 degrees as maximal shear,
- 10% maximal saturation and value change, to mimic different lighting,
- 80% chance of mosaic training (stitching multiple patches and their bounding boxes together) except for the last 10 epochs,
- 20% chance of mixup training (linearly combining images and their bounding boxes targets together)
- 40% chance of erasing random areas during training to force the model to learn more features.

Table 1: Summary of results

Method	Task	mAP50	CER
YOLOv12m	Detection	0.820	-
Stötzner et al.	Detection	0.602*	-
YOLOv12m	Detection + Classification + Transcription, all classes	0.340	0.494
YOLOv12m	Detection + Classification + Transcription, 251 classes	0.531	0.465
YOLOv12m	Detection + Classification + Transcription, 151 classes	0.676	0.493
Dencker et al.	Detection + Classification, 186 classes	0.656**	-
Simmons	Transcription (on Lineart)	-	0.354***

* Experiment on 512×512 patches.

** Experiment on photos.

*** Experiment on Lineart drawings.

4.4 Evaluation criteria.

We propose two evaluation criteria.

- **mAP50** for Detection or Detection and Classification task. This is one of the standard criteria used by bounding box detection tasks.

Before passing to evaluation, we perform NMS with IoU threshold of 0.5, and only keep predictions with confidence > 0.1 .

Note that when we regroup the rarest 200 classes, we also choose to regroup their true label. For instance, if the true label is OTHER and the model predicts OTHER, we choose to consider the prediction to be correct.

- **Character Error Rate (CER)** for transcription task. This is defined as

$$\text{CER} = \frac{\text{EditDistance}(\text{predicted}, \text{ground truth})}{\text{LEN}(\text{ground truth})}$$

where the EditDistance is the minimal number of insertions, deletions and substitutions needed to turn the first string into the second one.

We will report the mean CER on a subset (124 images) of the validation set (214 images): any validation image that has non-orthodox character names or has multiple columns does not participate in this evaluation.

5 Results

The principal results are summarized in Table 1.

5.1 Detection task.

We achieved 80.2% mAP50 on the Detection task. This is a significant improvement over the 60.2% mAP50 in Stötzner et al.[7]. Furthermore, our experiment is done on the whole tablet, and theirs only on the easier setting of 512×512 patches.

5.2 Detection and Classification task.

With all classes (including rare classes that are sometimes not even seen in the training set), the mAP50 is only 34.0%. However, if we only want to detect the most frequent 151 classes, the mAP rises to 67.6%, comparable to the complex pipeline built by Dencker et al. [7].

Of course, these results are not totally comparable by number, since the dataset is not the same (MSII renderings for us, photos for Dencker et al.).

5.3 Transcription task.

Using all classes, using only 251 classes and using 151 classes respectively give 49.4%, 46.5% and 49.3% CER. This is not quite satisfactory: as a reference, the end-to-end transcription system of Simmons [4] has only 35.4% CER on Lineart drawings.

However, we can still notice the following fact: adding a hyperparameter deciding how many signs to keep is useful. By abandoning the capacity to read the rarest signs, the overall robustness can increase.

5.4 Discussions.

5.4.1 Ablation studies.

We did not have the time to perform extensive ablation studies on all the parameters of the YOLO model. Our limited tests can only suggest the following facts: (1) that YOLOv12m is not so different from the older YOLOv8m [5] in terms of final results; they only differ in convergence speed; (2) that some of the data augmentation parameters do not have much impact on the final results; they only impact convergence speed. Further experiments are needed to fully confirm these claims.

5.4.2 Challenges that hinder performance.

We identify the following common causes of failure:

- The model struggles particularly on small or distorted signs at the boundaries of tablets, which

are very difficult to detect correctly using only frontal view.

- For composite signs that contain two sub-signs, the model often predicts the composite sign **and** one of the sub-signs. We have removed this case by only keeping the composite sign, but this is not always correct.
- A lot of signs on the tablets are **NOT** annotated by the MaiCuBeDa dataset. This, of course, lead to less robust training.

6 Further discussions

6.1 Failed attempts.

In order to test the possibility of directly using a pretrained Visual Language Model, we spent a lot of time (more than 3 weeks) doing the following experiments, which did not succeed:

- We fine-tune the newly-released Qwen3-VL-2B-Instruct [8] model end-to-end on the photographs in the SumTabletsPhoto dataset provided by Simmons [4]. We want to test if the model can learn to directly provide a sequential transcription of all signs on the photo. The model, however, does not converge and only outputs repetitive signs.
- Qwen3-VL-2B-Instruct claims to have good capacity on OCR bounding box predictions. We thus train it end-to-end on the bounding box annotations. The model, however, does not converge and only outputs randomly guessed signs and bounding boxes.
- To see if the Qwen3 model can be trained to understand Sumerian signs at all, we train it to simply classify cropped signs. Even this does not work and the model only outputs randomly guessed signs.

We can but only conclude that the existing Visual Language Models like Qwen3 do not have any visual knowledge of Sumerian cuneiform signs, and our small quantity of data is not sufficient for effective training.

6.2 Possible improvements and road-maps.

- **Training on real photos.** In our experiences, we only used the dataset of MSII-rendered images: although [7] transformed the bounding boxes on MSII images onto real photos by manually marking the corresponding regions, they only published the tool and not the resulting bounding boxes on the real photos.

As a consequence, we observe that our trained YOLO model does not generalize well enough to identify signs on real photos. Using a more specific pipeline or obtaining a photo-bounding box dataset is needed.

- **Creating synthetic data.** It is common practice for complex OCR systems to pretrain on large-scale synthetic data. The is the case for many popular OCR systems, including Donut [9], PP-OCR [10] and TrOCR [11]

- , which all use synthetic data to enhance the ability to read computer fonts.

For us, it would be possible to make a larger synthetic dataset using various cuneiform fonts, but this could not totally mimic real carved signs on a tablet. Besides, training on synthetic datasets is very costly in terms of time. More intelligent solutions for data augmentation may be needed.

- **Towards transliteration and language models.** For the time being, all our experiments use the Unicode character as the ground truth label. However, in Sumerian, a character may have multiple readings and pronunciations, just like Chinese. We need a language model to correctly capture the reading of a sign in context.

In this direction, Simmons [12] proposes a sign reading framework, which for each Unicode string returns the transliteration (reading) of all the signs. It would be possible to combine the detection framework with this transliteration framework to make a fully image-to-transliteration Sumerian reading system.

However, on the other hand, it would have also been interesting to see if the language model (i.e. contextual linguistic information) can enhance the **detection** of the signs. A more intelligent conception might be needed to achieve this.

7 Conclusions

We demonstrated that modern object detection frameworks can be effectively adapted for Sumerian OCR, treating each cuneiform sign as a detectable object. Our YOLOv12-based pipeline achieves strong performance in both sign detection and classification, while a reading order module enables sequential transcription into Unicode text. Despite challenges such as limited real-photo datasets and rare sign classes, our approach provides a practical and scalable solution for automated cuneiform reading. Our work can be improved by extending the system to real photographs, integrating language modeling for contextual reading, and generating synthetic data to improve rare sign recognition.

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